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Rock Piles as Kōura Habitat

Habitat enhancement for a native freshwater crayfish threatened by an invasive predator: a field trial of stone pile structures in an Aotearoa New Zealand lake

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Abstract

1. Kōura (*Paranephrops planifrons*), Aotearoa New Zealand’s endemic freshwater crayfish, have declined by up to 96 % in Lake Rotoiti (Te Arawa Lakes, Rotorua), driven by predation from the invasive brown bullhead catfish (*Ameiurus nebulosus*), thermal stratification, and invasive macrophytes that compress suitable habitat into the shallow littoral zone. Because these stressors cannot be removed at the lake scale, adding structural refuge may be one of few practical management options.
2. We tested whether stone pile structures deployed in the littoral zone of Lake Rotoiti could provide structural refuge for kōura from predation. Kōura and catfish were monitored at five stone piles, five control, and five paired natural rocky reference sites for 18 months using a quasi before-after control-impact (quasi-BACI) design, with kōura occupancy modelled using a Bayesian framework that incorporated detection probabilities from baited traps and fyke nets.
3. Stone pile construction produced no statistically significant change in kōura abundance, biomass, or occupancy relative to controls, but effects consistently trended positive across models; a simulation-based power analysis showed the study was underpowered to detect the observed effect size, given high pre-construction occupancy (81–89 %) at all site types and only five site pairs.
4. Kōura use of stone piles was strongly seasonal: catch rates at stone pile and control sites fell in winter as kōura withdrew to deeper water, while reference sites remained stable. Catfish showed no comparable attraction to stone piles in any season or model.
5. *Synthesis and applications.* Stone piles show early, consistent evidence of benefiting a threatened native crayfish without benefiting its invasive predator and are most valuable when habitat compression and predation pressure peak together in summer and autumn, conditions expected to intensify under climate warming. This study establishes a tested, scalable construction and consenting methodology that lake managers can deploy as an interim refuge-based tool while catfish eradication remains unachievable at the lake scale; we recommend continued monitoring beyond three years and trialling connected, corridor-style structures rather than isolated piles.

Introduction

Freshwater biodiversity is declining faster than any other biome, with one quarter of freshwater fauna now threatened with extinction (Sayer et al., 2025). Among the most endangered are

1 freshwater crayfish, with 32 % of species assessed as threatened globally due to habitat loss,
2 invasive species, and declining water quality (Richman et al., 2015). As keystone species and
3 ecosystem engineers, crayfish play crucial roles in nutrient cycling, bioturbation, and food web
4 dynamics, and their loss can have cascading consequences for freshwater ecosystems (Collier
5 et al., 1997; Momot, 1995; Parkyn et al., 2001). In Aotearoa New Zealand, these pressures
6 converge on kōura (*Paranephrops planifrons* White, 1842), an endemic freshwater crayfish now
7 facing rapid population decline in lakes that have long supported some of the largest customary
8 kōura fisheries in the country (Hiroa, 1921; Kusabs & Quinn, 2009).

9 Kōura populations in Lake Rotoiti, located in the Rotorua Te Arawa Lakes region of the North
10 Island of Aotearoa New Zealand (hereafter Aotearoa), have declined by up to 96 % over the
11 past two decades (Kusabs et al., 2026). This collapse has virtually ended customary harvest
12 practices for Te Arawa iwi (Māori tribal group) who are hunga tiaki (traditional custodians)
13 of the lake (Kusabs, Hicks, et al., 2015). The decline in kōura numbers is driven by multiple
14 interacting stressors that compound over time in different parts of the lake (Kusabs et al., 2026;
15 Lee et al., 2025). The invasive brown bullhead catfish (*Ameiurus nebulosus* Lesueur, 1819,
16 hereafter catfish), confirmed in Lake Rotoiti in 2016 (Francis, 2019; Grayling, 2020), preys
17 directly on kōura and suppresses their foraging through non-consumptive ‘fear’ effects (Raven
18 et al., 2026a). Predation pressure from catfish is compounded by additional stressors that
19 constrain where in the lake kōura can seek refuge from this invasive predator (Kusabs et al.,
20 2026).

21 During the day, kōura shelter in complex rocky habitat and in deeper water to avoid predation
22 (Jowett et al., 2008; Kusabs, Quinn, et al., 2015), moving out of both at night to forage in
23 the productive littoral zone (Devcich, 1979). The importance of deep-water refuge is shown in
24 Lake Taupō, where kōura persist at a depth even in the presence of catfish because the lake
25 remains oxygenated below the thermocline (Kusabs & Taiaroa, 2015; Verburg & Albert, 2019).
26 In Lake Rotoiti, this deep refuge becomes anoxic during summer stratification and unavailable
27 for kōura, compressing the zone of liveable habitat into the shallow littoral margins (Kusabs
28 et al., 2026), precisely where catfish are most active (Blanchfield et al., 2026; Dedual, 2002).
29 Dense beds of invasive macrophytes further restrict kōura movement through this compressed
30 zone and intensify oxygen depletion during die-off events (Hamilton et al., 2005; Kusabs &
31 Quinn, 2009; Vincent et al., 1984). Catfish pressure and thermal stratification are expected
32 to only strengthen with predicted climate scenarios (Lee et al., 2025; Woolway et al., 2021),
33 making kōura habitat compression and predation pressure progressively more severe.

34 The compression of suitable habitat and subsequent concentration of kōura into the littoral

1 zone makes the quality of nearshore lacustrine habitats increasingly important for remaining
2 populations. In stream environments, kōura are positively associated with structurally complex
3 habitats including bank undercuts, woody debris, leaf litter, and coarse substrates (Jowett
4 et al., 2008; Parkyn et al., 2009; Parkyn & Collier, 2004; Usio & Townsend, 2000). In lake
5 environments, benthic substrate characteristics are a stronger predictor of kōura distribution
6 than nutrient enrichment or predator abundance (Kusabs, Quinn, et al., 2015), and kōura
7 are positively associated with coarse substrates and riparian vegetation (Raven et al., 2026b).
8 Coarse rocky substrate provides refugia from predators and shelter during moulting (Streissl
9 & Hödl, 2002). Experimental studies confirm that shelter availability directly reduces catfish
10 predation on crayfish (Adams, 2007; Zhao et al., 2024), and kōura increase refuge use in
11 the presence of catfish compared to native predators (Raven et al., 2026a), emphasising the
12 importance of shelter availability for kōura populations. Shelter availability is likely a limiting
13 factor for kōura in Lake Rotoiti, where the majority of the shoreline is sandy with dense beds
14 of introduced macrophytes and little rocky structure. Adding rocky substrate to soft-bottomed
15 lake areas has been shown to significantly increase crayfish abundance (Johnsen & Taugbøl,
16 2008), and increasing structural complexity in streams improves crayfish restocking success
17 (Huolila et al., 1997). Deploying habitat enhancement structures that add structural complexity
18 in the littoral zone could potentially be a practical way to restore kōura populations in Lake
19 Rotoiti.

20 In this study habitat enhancement structures in the shape of stone piles were deployed along
21 the shoreline of Lake Rotoiti to investigate the ability of the structures in providing additional
22 habitat for kōura to live and hide from predators. Kōura presence, abundance, and biomass
23 were monitored at stone pile, control, and reference sites over 18 months. We predicted that
24 kōura occupancy, abundance and biomass would increase more rapidly at the stone pile sites
25 than at control sites, as intense catfish predation pressure in the littoral zone was expected
26 to compel kōura actively toward available refuge structures. We further predicted that kōura
27 abundance would continue to increase progressively after construction, at least within the first
28 18 months, and that catfish would instead show seasonal patterns of shoreline use rather than
29 a construction-driven trend (Barnes, 1996; Blanchfield et al., 2026; Dedual, 2002). Finally, we
30 predicted that stone pile sites would progressively converge toward kōura densities observed at
31 natural rocky reference sites, and that catfish would not exhibit a similar pattern.

1 **Materials and Methods**

2 **Study location and construction**

3 Lake Rotoiti (-38.03, 176.42) is a warm monomictic mesotrophic lake located in the Rotorua
4 Te Arawa Lakes region of the North Island of Aotearoa (34 km², maximum depth 128 m).
5 The lake stratifies for approximately nine months a year (Figure 7), with pronounced thermal
6 stratification compressing oxygenated habitat into the shallow littoral zone during summer
7 and autumn. Since 2008, nutrient-rich inflows from Lake Rotorua have been diverted toward
8 the Kaituna River via the Ōhau Channel diversion wall to improve water quality in Lake
9 Rotoiti (Hamilton et al., 2005; Prentice et al., 2026). Lake level is regulated by a weir at the
10 outlet of the Kaituna River, providing a stable and controlled water level throughout the year.
11 Catfish are present throughout the lake but occur in particularly high concentrations in Te
12 Weta Bay on the Northwest side of the lake (Francis, 2019). Five stone pile sites were selected
13 alongside the lake shoreline based on accessibility and landowner approval (Figure 1). At each
14 stone pile site, a paired control site was established 50 m along the shoreline, matched for
15 habitat characteristics e.g., water depth, substrate type and vegetation cover. Five naturally
16 occurring rocky shoreline locations, with each two monitoring sites 50 m apart were selected
17 as reference. In total, 20 sites of each 150 m² (15 × 10 m) were monitored seasonally over 18
18 months. Physical characteristics of all sites are summarised in Table 1. Baseline monitoring was
19 conducted in February 2025 prior to stone pile deployment, which occurred the following week.
20 Stone piles were constructed from locally sourced quarry rock (Manawahe Stone and Quarry,
21 ~25 km from the lake) with a diameter ranging from 85 - 315 mm (mean = 173 mm). At each
22 of the five stone pile sites, three stone piles were arranged horizontally along the shoreline by
23 hand from the shore. Stone pile design was informed by mesocosm experiments demonstrating
24 kōura preference for enclosed stone structures (unpublished data). Stone piles were positioned
25 approximately 4.3 m from shore at a mean depth of 0.49 m and with a mean area of 4.1 m².

26 **Monitoring**

27 Physical, chemical, and biological parameters were recorded at each site during each monitoring
1 round. Static site characteristics including substrate composition, slope to depth contour,
2 riparian vegetation, overhanging tree cover, and wood cover were visually estimated during
3 baseline monitoring only, as these were not expected to change substantially over the monitoring
4 period. Substrate composition was visually estimated by a single observer walking around

each site and using an underwater viewing chamber, categorising bedrock, boulders (>256 mm), cobble (64–256 mm), gravel (2–64 mm), sand (0.063–2 mm), mud (<0.063 mm), and organic matter (Wentworth, 1922). A substrate index, that indicates suitability for kōura, was calculated from these estimates (Jowett et al., 2008; Jowett & Richardson, 1990):

$$\begin{aligned} \text{Substrate index} = & 0.08 \times \text{bedrock} + 0.07 \times \text{boulders} \\ & + 0.06 \times \text{cobble} + 0.04 \times \text{gravel} + 0.03 \times \text{sand} \\ & + 0.02 \times \text{organic matter} + 0.01 \times \text{mud} \end{aligned}$$

Coarser substrates were assigned higher weights as they provide greater interstitial pore space and stable structures used by kōura (Nyström et al., 2006; Usio & Townsend, 2000).

After stone pile construction, measurements of the size and dimensions of each structure were taken. The post-construction substrate index for stone pile sites was recalculated using each site’s combined stone-pile footprint area, expressed as a percentage of the 150 m² site area and treated as 80 % cobble and 20 % boulders displacing an equivalent area of sand, to reflect their contribution to substrate complexity. Slope was assessed as the horizontal distance from the shoreline to the 20 m depth contour, derived from bathymetric maps and measured in a projected coordinate system (NZTM2000; EPSG:2193).

Seasonal variable parameters were recorded at every monitoring round and included aquatic vegetation cover (emergent and submerged macrophytes). Chemical parameters included water temperature (°C), dissolved oxygen (mg L⁻¹), and conductivity (μS cm⁻¹) were measured using a YSI ProSolo meter (Pro2030, YSI Inc., Yellow Springs, OH, USA). pH was measured using a handheld pH meter (EC-PCTestr35, Eutech Instruments).

Kōura and fish were sampled every monitoring round using two single-winged, double-throated fyke nets (600 mm high D-shaped entrance hoop) and two collapsible bait traps (480 × 250 mm; two 50 mm entrance openings) baited with catfish flesh (Joy et al., 2013; MacNeil et al., 2025). Fyke nets were placed perpendicular to the shoreline outside of the stone pile structures or at 10 m intervals, with bait traps positioned between them. All equipment was deployed simultaneously at each site for a 24-h period. Captured kōura were measured for orbital carapace length (OCL) using a digital calliper (0.01 mm precision; 150 mm) and for wet mass using a digital scale (± 1 g; 5 kg capacity, Anko Brand, Kmart Group, Mulgrave, VIC, Australia). Catfish and other fish species, were recorded for abundance, measured for total length, and weighed for wet mass. Catch per unit effort (CPUE) and biomass per unit effort (BPUE) were calculated for all kōura and fish based on the number of nets and traps

4 deployed at each site per monitoring round, standardised by soak time (Kusabs, Quinn, et
5 al., 2015). Kōura weights were unavailable for 36 individuals because of scale failure and were
6 therefore estimated from a log $-\log$ length–weight regression fitted to individuals with both
7 measurements ($n = 840$), following standard approaches for weight length relationships in
8 aquatic organisms Figure 8 (Froese, 2006). Predicted values were back-transformed with a
9 lognormal bias correction factor $10^{0.5\sigma^2}$ (Quinn & Keough, 2002).

$$\log_{10}(\text{Weight [g]}) = -2.67 + 2.684 \times \log_{10}(\text{OCL length [mm]})$$

10 Data analysis

11 Data analysis was a three-part process, the first assessed baseline equivalence between stone
12 pile, control, and reference sites before construction, the second assessed the impact of stone
13 pile construction and after-construction trajectory on kōura and catfish occupancy, abundance,
14 and biomass, and the third assessed whether stone pile sites converged toward reference sites
15 and if reference sites remained stable over time. All analyses were conducted in R (version
16 4.6.1; R Core Team (2026)) using the glmmTMB and emmeans packages (Brooks et al., 2017;
17 Lenth & Piaskowski, 2017).

18 Covariate selection for all models followed a consistent procedure. Corrected Akaike Information
19 Criterion (AICc) values for each base model were compared against models with additional
20 covariates (temperature, season, substrate index, maximum effective fetch, or wave energy
21 proxy) added at a time, and a covariate was retained only when it reduced AICc by more than
22 two units relative to base model (Burnham & Anderson, 2002). Robustness and sensitivity for
23 all models were assessed by evaluating uniform distribution, dispersion and outlier rate using
24 the DHARMA package (Hartig, 2016). Across the full set of models, the Benjamini–Hochberg
25 correction method was applied to control the false discovery rate within each of the three
26 analysis components.

27 Site variation before stone pile construction was assessed using paired Wilcoxon signed-rank
28 tests to compare stone pile sites with their paired control sites, and unpaired Wilcoxon rank-sum
29 tests to compare stone pile sites with reference sites. This was done for response variables
30 (water quality, substrate, and kōura and catfish CPUE and BPUE) to confirm that the stone
31 pile and control sites were equivalent before construction, and to characterise the difference
32 between stone pile and reference sites. To further assess baseline equivalence, generalised

linear models (GLMs) were fitted to test whether kōura and catfish CPUE and BPUE differed between site type (stone pile / control) and between the two reference sites (reference 1 / reference 2) using a Tweedie family with a log link. Following the covariate selection procedure, temperature was included in the catfish models comparing reference sites, as the base model failed to converge and temperature was required for numerical stability.

The second component of the analysis tested construction impact relative to control sites using a quasi-before-after, control-impact (quasi-BACI) design (Underwood, 1994), and separately tested after-construction trajectory of kōura and catfish populations. The design is quasi-BACI because only a single pre-construction monitoring round was possible. The quasi-BACI design compared one pre-construction monitoring round with six post-construction monitoring rounds using generalised linear mixed effects models (GLMMs) to test the effect of stone pile construction on kōura and catfish occupancy, abundance, and biomass. Time period (before / after) and site type (stone pile / control) were included as an interaction term, monitoring round as a covariate and site nested within control-impact pair as a random effect. Occupancy was modelled using a binomial GLMM, and abundance and biomass using a Tweedie GLMM with a log link; fetch was retained as a covariate in the catfish model. The post-construction trajectory models used the same response variables, families, covariate, and random effect structure, but with site type and monitoring round included as an interaction term in place of the before / after comparison.

The third component of the analysis assessed whether stone pile sites converged toward reference sites over time, and whether reference sites themselves remained stable. Kōura and catfish abundance and biomass were tested using GLMMs with site type (stone pile / reference) and monitoring round included as an interaction term, and site number as a random effect, using a Tweedie family with log link; temperature was retained as a covariate in the catfish model. Reference site stability was tested in GLMMs with site type (reference 1 / reference 2) and monitoring round as an interaction term, and site nested within pair as a random effect, again using Tweedie family with log link; season was retained as a covariate in the kōura model and temperature was retained as a covariate in the catfish model.

To complement the GLMM abundance analysis, kōura site-level occupancy was estimated using a two-method occupancy model fitted within a Bayesian framework. Because bait traps and fyke nets provide independent and simultaneous detection opportunities at each site, true site occupancy could be separated from imperfect and method-specific detection probabilities. A custom Stan observation model was implemented using the brms package (Bürkner, 2017, 2018) using cmdstanr (Gabry et al., 2025) to estimate true occupancy and detection probabilities

for baited traps and fyke nets via logit links. Standardised water temperature was included as the covariate for detection probability. Three occupancy sub-models were fitted. A full model across all monitoring rounds served as baseline comparison. A summer-only model and a warm-season model (summer and autumn combined) were fitted to focus on the period of highest thermal stratification, when kōura are known to disproportionately occupy nearshore littoral habitats (Kusabs et al., 2026) and when catfish predation pressure is highest in the areas where the stone piles are located (Blanchfield et al., 2026). The warm-season model maximises ecological relevance, while retaining a tractable formula with season as an additive covariate. All models were run with four chains of 4000 iterations (1500 warmup) and convergence was assessed using R values (threshold < 1.01) and bulk effective sample sizes (threshold > 400) (Vehtari et al., 2021). Divergent transitions were rare in all models (9 of 10,000 post-warmup draws) and considered negligible given the otherwise strong convergence diagnostics.

A simulation-based power analysis was conducted to quantify the sample size required for statistical detection of the BACI effect under the observed effect size and variance structure, and to put the non-significant results in perspective relative to study design rather than true absence of effect. For each analysis the occupancy and abundance, datasets were simulated across a grid of scenarios varying in number of site pairs (3-30) and after-period monitoring rounds (2-12). For the occupancy BACI, simulated binary detection data were generated using values from the warm-season occupancy model posterior means, and each simulated dataset was analysed using a logistic GLMM (lme4; Bates et al. (2015)). For the CPUE BACI, simulated abundance data were generated using fixed effects and variance components from the kōura abundance BACI model, using a log-normal approximation to the Tweedie distribution, and each dataset was analysed using a linear mixed model fitted to a log-transformed CPUE. Statistical power was estimated as the proportion of simulations (300 per scenario) in which the model's 95 % confidence interval for the BACI coefficient excluded zero in the positive direction.

Results

Kōura presence was detected at stone pile sites within three months of construction, with occupancy and abundance trending consistently higher at stone pile sites than paired control sites throughout the 18-month post-construction period, though this difference did not reach statistical significance. Results across all models and response variables are summarised in Figure 2 and ?@tbl-models-summary-full-suppl.

4 Baseline equivalence

Pre-construction, kōura abundance and biomass did not differ significantly between stone pile and paired control sites (?@tbl-baseline-combined-suppl), confirming baseline equivalences ($p = 0.783$ and $p = 0.571$). Kōura abundance ($p = 0.968$) and biomass ($p = 0.968$) did not differ between reference sites. Catfish abundance ($p = 0.004$) and biomass ($p = 0.002$) likewise did not differ between stone pile and control sites pre-construction. However, catfish abundance and biomass were significantly higher at Reference 1 than Reference 2 ($p = 0.004$ and $p = 0.002$, respectively), despite both sites having comparable rocky substrate. This suggests catfish distribution varies considerably between nearby sites with similar habitat characteristics.

13 Construction impact (BACI) and post-construction trajectory

Stone pile construction produced no detectable step-change in kōura abundance, biomass, or occupancy relative to paired control sites over the 18-month monitoring period following deployment (all $p_{\text{adj}} = 0.858$) (Figure 3). A matched-season comparison, comparing Summer 2025 (before construction) with Summer 2026 (one year after construction), produced the same conclusion (kōura abundance: $p = 0.582$). Post-construction trajectories did not diverge significantly between stone pile and control sites (all $p_{\text{adj}} = 0.215$). Kōura abundance declined significantly across post-construction rounds at control and stone pile sites ($\beta = -0.41$, $p = 0.004$), with the sharpest decline observed in Winter 2026 (Figure 4). Stone pile construction produced no detectable step-change or trajectory divergence in catfish abundance, biomass, or occupancy relative to control sites across any model or time period (Figure 2).

Kōura occupancy was high across all site types before construction, 81–89 % in summer (?@tbl-psi-warm-suppl). After construction, occupancy increased marginally at all sites, 89–94 % in summer, thereby limiting our ability to detect a treatment effect. The BACI contrast to compare how much stone pile sites changed relative to control sites was slightly positive, but the credible intervals spanned zero with wide uncertainty (Figure 5), indicating no detectable additional increase at stone pile sites. However, this result likely reflects a measured ceiling rather than biological response. After-treatment contrast showed no significant difference between site types, but the direction of the effect was consistent with early habitat use: stone pile sites showed a positive contrast relative to control sites, while control sites showed a negative contrast relative to reference sites, and stone pile sites were near zero relative to reference sites (Figure 5).

5 The non-significant BACI results should be interpreted in the context of statistical power. For
6 occupancy, power was near zero across all simulated scenarios (maximum 8%), reflecting the
7 ceiling effect leaving no statistical room for a BACI signal (Figure 9). Post-construction power
8 analysis showed that achieving 80 % power with the current five site pairs would require an
9 effect size of approximately 1.8 log-units, almost 6 times the current observed effect (Figure 6).
10 If the estimated positive trajectory of +0.24 log-units per seasonal round continues, the effect
11 size may reach this detection threshold within approximately 6 additional monitoring rounds
12 (~1.5 years).

13 **Convergence toward reference sites and reference stability**

14 Stone pile sites maintained a consistently lower kōura abundance than reference sites throughout
15 the post-construction period, with no significant change in this gap over time ($\beta = -0.09$, $p =$
16 0.429 ; Figure 10). Kōura abundance declined across monitoring rounds at both stone pile and
17 reference sites ($\beta = -0.09$, $p = 0.107$). The Winter 2026 monitoring round showed a pronounced
18 decline at stone pile sites (CPUE 0.7) while reference sites maintained substantially higher
19 abundance (CPUE 1.7; Figure 4), a within-round contrast not captured by the linear trajectory
20 model. Water temperature at Winter 2026 was approximately 1.5°C lower than Winter 2025 but
21 consistent across site types, ruling out temperature as a site-specific driver. Visual inspection
22 of stone pile surfaces during Winter 2026 monitoring revealed substantially greater filamentous
23 algal accumulation compared to earlier rounds (?@fig-stone-piles). Catfish abundance and
24 biomass showed no significant convergence or divergence between stone pile and reference sites
25 over time (Figure 2).

26 **Discussion**

27 Stone pile structures were deployed in the littoral zone of Lake Rotoiti to test whether physical
28 refuge could reduce predation pressure on kōura by invasive catfish. Stone piles produced no
29 statistically significant increase in kōura abundance, biomass, or occupancy relative to paired
1 control sites, but the direction of effect was consistently positive across every response variable
2 and model, and occupancy at stone pile sites converged with occupancy at natural rocky
3 reference sites by the first post-construction summer. Catfish showed no comparable response
4 to stone pile presence in any model, at any site type, or in any season, indicating that stone
5 piles selectively benefit kōura without providing equivalent refuge or foraging opportunity for

6 this invasive predator. A pronounced decline in kōura catch rates at stone pile and control sites
7 in Winter 2026, evidently driven by seasonal redistribution to deeper water rather than habitat
8 abandonment, showed that structural refuge value is itself seasonal and concentrated in the
9 warmer months when catfish predation pressure and habitat compression peak. Together, these
10 results and a formal power analysis indicate that the absence of statistical significance most
11 plausibly reflects limited study duration and sample size rather than genuine ineffectiveness of
12 the intervention.

13 **Construction effect and statistical power**

14 Kōura catch rates were highly variable across all site types throughout the monitoring period,
15 with wide confidence intervals, likely reflecting genuine biological variability in kōura distribution.
16 This variability may reflect seasonal activity patterns, detection probability changes across
17 monitoring rounds, and natural heterogeneity in littoral zone use, rather than stable site
18 associations. Despite this variability, the Bayesian occupancy model revealed a consistent
19 positive directional trend, with stone pile sites showing higher occupancy than control sites,
20 and occupancy comparable to natural rocky reference sites across both summer and autumn
21 monitoring rounds. Whether this reflects active refuge-seeking driven by catfish predation
22 pressure or gradual colonisation and population expansion into newly available habitats cannot
23 be determined from the current data, but the direction of the response was consistent with
24 the purpose of the structures, yet no statistically significant effect was detected within the
25 18-month monitoring period.

26 The absence of a significant effect is most likely explained by a combination of monitoring
27 duration, statistical power, and system dynamics rather than ineffectiveness of the structures.
28 Habitat establishment for freshwater crayfish can be inherently slow. Johnsen & Taugbøl (2008)
29 reported that crayfish abundance at a stone addition site took 13 years to reach reference
30 levels, and monitoring periods shorter than two years are often insufficient to detect significant
31 responses to habitat modification (Pilotto et al., 2018; Ruhí et al., 2012; Testa et al., 2011).
32 Following reconstruction of the Lake Rotorua lakefront, which is located in close proximity
33 to our study site, kōura showed no meaningful recolonisation after 28 months (Kusabs, 2023).
1 They attributed slow recolonisation to persistent catfish predation pressure and distance from
2 source population, consistent with time scales observed here. The power analysis indicates that
3 five site pairs are insufficient to reliably detect effects of the magnitude observed here within
4 18 months, establishing a minimum design threshold: future stone pile monitoring programmes

5 should plan for at least three years of monitoring and expect high pre-construction occupancy,
6 leaving little statistical room for a measurable treatment effect. The simultaneous elevation of
7 kōura CPUE across all site types in Autumn 2025 further complicated detection of an early
8 treatment effect in the BACI contrast.

9 Features of the system itself may also contribute to the weak response. Kōura abundance
10 is highest in lakes without catfish ([Raven et al., 2026b](#)), suggesting predation pressure may
11 suppress kōura populations well below the carrying capacity of existing habitat. If so, structural
12 habitat availability may not currently be the primary limiting factor, and a weak or delayed
13 response to habitat addition would be expected. Equally, if kōura are already at carrying
14 capacity of existing rocky habitat, stone piles add new capacity that will only be filled gradually
15 as the population expands, which is unlikely to be detectable within 18 months. In either
16 case, a delayed rather than absent population response would be expected. Isolated stone pile
17 structures may also be difficult for kōura to locate, since distance between isolated habitat
18 patches is a key determinant of colonisation success ([Hanski, 1998](#)). Each of these explanations
19 point toward a delayed, rather than absent, population response, and continued monitoring
20 would be the appropriate next step rather than a reassessment of the intervention itself.

21 **Seasonal dynamics: the Winter 2026 decline**

22 Kōura CPUE at stone pile and control sites fell sharply in Winter 2026, returning to levels
23 close to the pre-deployment baseline from Summer 2025, while reference sites maintained
24 substantially higher abundance. This pattern is consistent with known seasonal migration
25 behaviour across lake depth profiles and kōura activity rates during cooler water temperatures
26 ([Devcich, 1979](#)). When lake thermal stratification breaks down in winter, the hypolimnion
27 re-oxygenates and deep-water habitat becomes accessible, allowing kōura to redistribute along
28 the depth gradient ([Devcich, 1979](#)). Reduced activity at low water temperatures independently
29 lowers the probability of trap capture ([Devcich, 1979](#)), and although Winter 2026 was 1.5 °C
30 lower than Winter 2025, this difference was consistent across all site types and therefore cannot
31 explain why reference sites were substantially more resilient to the winter decline than the
32 stone pile sites.

1 Natural rocky reference sites in this study represent the ecological target for stone pile deploy-
2 ment. Reference sites had established kōura populations, connectivity to deeper waters, and
3 extensive refuge habitat, allowing kōura to maintain littoral presence through winter or return
4 rapidly from deeper refuges as conditions fluctuated. Stone pile sites are isolated structures with

5 less connectivity to deep and extensive refuge habitat, meaning kōura that withdrew in winter
6 had no quick pathway back to the structures. This difference likely explains why stone pile sites
7 maintained consistently lower kōura abundance than reference sites throughout the monitoring
8 period, with no significant convergence toward reference levels. However, kōura occupancy at
9 stone pile sites was comparable to that at reference sites after construction, suggesting that
10 kōura found and used the structures at similar frequency to natural rocky habitat but at lower
11 densities. This distinction between occupancy and abundance likely reflects the early stage of
12 colonisation and the small structural footprint of stone piles relative to the extensive natural
13 rocky habitat at reference sites. These differences are expected to narrow as biofilm succession
14 matures and kōura accumulate at the structures over time.

15 Extensive filamentous algae growth was observed on the stone pile structures in Winter 2026
16 but not to the same degree in Winter 2025 (?@fig-stone-piles), consistent with the seasonal
17 pattern typical for thermally stratified lakes, where winter mixing raises nutrient availability
18 while cooler water suppresses invertebrate grazing, allowing algal mats to accumulate on hard
19 surfaces (Bonacina et al., 2023; Wetzel, 2001). The freshly quarried piles were free from biofilm
20 at deployment and by Winter 2026 supported a diverse meiofaunal assemblage, indicating
21 active succession rather than habitat degradation; algal cover would be expected to decline as
22 grazer activity resumes in spring and summer. Despite the near-zero trap catches in Winter
23 2026, night spotlighting at two stone pile sites in August 2026 directly observed kōura sheltering
24 within the structures, confirming that low winter catch rates reflect reduced detectability rather
25 than abandonment. Together, these results indicate that stone pile refuge value is inherently
26 seasonal — greatest in summer and autumn, when thermal stratification compresses habitat
27 into the littoral zone and catfish predation peaks, and lowest in winter, when kōura redistribute
28 to deeper water regardless of structure type.

29 **Catfish: no evidence of attraction to stone piles**

30 Catfish showed their seasonal patterns in littoral zone use, but these were unrelated to stone pile
31 presence. Stone pile construction produced no evidence of catfish attraction or accumulation.
1 Even if catfish were to use stone piles to some degree, the ecological significance of this would
2 be limited given the extensive availability of existing suitable habitat for catfish in Lake
3 Rotoiti, particularly invasive macrophyte beds (Barnes & Hicks, 2003). The single-site increase
4 in catfish CPUE during Summer 2026 was mirrored by an increase at reference sites and
5 was consistent with lake-wide seasonal redistribution into warm littoral waters rather than

any stone-pile-specific event (Blanchfield et al., 2026). Overall catfish catches remained low throughout the study relative to targeted removal netting in Te Weta Bay, where highest catfish densities are concentrated (Francis, 2019). In autumn, catfish shift to mid-water movement across the lake basin remaining above 15 m depth (Blanchfield et al., 2026), reducing predation pressure on kōura in the littoral zone. This absence of attraction is an important management result in its own right: habitat enhancement can be targeted at a threatened native species without inadvertently improving conditions for its invasive predator.

Implications for restoration design and monitoring

Predator-prey interactions are shaped by habitat structure, which offers refuge to prey, influences how easily predators locate and capture prey, and alters the costs and benefits of hunting for a predator (Lennox et al., 2025). This framework applies directly to catfish-invaded lakes, where habitat addition offers a practical lever on that interaction, and rocky environments have been identified as natural multi-threat refugia, capable of simultaneously buffering predation, thermal extremes, and invasive species pressure across a wide range of taxa and ecosystems (Selwood & Zimmer, 2020). In Lake Rotoiti, catfish and thermal stratification — the two primary drivers of kōura decline — cannot be managed at the lake scale: eradication is not currently achievable, and targeted removal in high density areas reduces local pressure without eliminating the lake-wide population (Dedual, 2019). Emerging biotechnologies such as gene drives and host-specific biological control agents may offer future options (Kopf et al., 2019; Teem et al., 2020; Thresher et al., 2014), but these remain experimental, and thermal stratification will continue to intensify under climate warming (Lee et al., 2025; Woolway et al., 2021). Where the primary stressors cannot be removed, habitat addition becomes one of the few available management tools, and refugia become correspondingly more valuable (Selwood & Zimmer, 2020).

The stone piles deployed here were isolated structures, separated from each other and from deeper water by soft sandy substrate. Isolated habitat patches support fewer species and are more vulnerable to local extinction than connected habitats (Hanski, 1998), and the Winter 2026 decline is consistent with this limitation: stone piles lacked the connectivity to deep refuge that allowed reference sites to remain stable. Orienting future structures perpendicular to the shoreline, extending through macrophyte beds and into deeper water, could create corridors linking the littoral zone with deeper refugia, allowing kōura to move more freely between depth ranges as conditions change, as they do at reference sites. This design has not been tested

4 but is a promising direction for future deployments, with the addition of potential benefit of
5 suppressing macrophyte growth along the corridor.

6 The dense algal fouling observed in Winter 2026 also raises a placement question. In sheltered
7 bays, stone piles are protected from wave damage, but algal growth and sediment accumulation
8 could progressively reduce crevice accessibility and refugia value; in more exposed locations,
9 natural scouring could limit fouling, but conditions may be too disturbed to provide effective
10 shelter (Parkyn & Collier, 2004). This placement trade-off warrants direct investigation in
11 future deployments, and standardised photography of consistent stone faces each monitoring
12 visit would allow fouling development to be tracked alongside changes in kōura use.

13 Finally, the five stone pile structures deployed here cannot be expected to drive population-
14 level recovery across a lake of this size on their own. The spatial and temporal scale of the
15 ecosystem responses is typically greater than the scale of the interventions used to produce
16 them (Minns et al., 1996), and scaling from individual stone pile deployment to habitat addition
17 at population-relevant extents is a central challenge in restoration ecology (Ngwenya et al.,
18 2025).

19 Beyond the ecological results, this study establishes a practical foundation for scaling habitat
20 enhancement in these lakes: the approvals, consents, and construction methodology needed to
21 deploy stone piles have now been tested, with evidence that doing so is unlikely to meaningfully
22 benefit catfish on their own. These outcomes give lake managers a practical, tested option for
23 future large-scale deployment; continued monitoring for at least three years is recommended to
24 determine whether the Winter 2026 decline was a one-off event and whether kōura abundance
25 at stone pile sites continues to build in subsequent spring and summer seasons.

26 **Conclusion**

27 Stone pile deployment alone cannot reverse the trajectory of kōura decline in Lake Rotoiti,
28 but it directly addresses the current bottleneck: the shortage of structural refuge in a lake
29 where soft sediment dominates the shorelines and where the two primary stressors, catfish
1 predation and thermal habitat compression, cannot be controlled at the lake scale. Both
2 stressors are expected to intensify under continued climate warming (Lee et al., 2025; Woolway
3 et al., 2021), narrowing the band of habitat available to kōura and increasing, rather than
4 diminishing, the value of structural refugia over time. Establishing refugia before populations
5 decline further may be more effective than waiting until source populations fall below critical

6 densities for recolonisation (Kusabs et al., 2026; Morelli et al., 2016). The consistent positive
7 directional trend in occupancy, the rapid and clearly seasonal use of the structures, and the
8 developing biological community on the stone pile surfaces together suggest the foundations for
9 a population-level response by kōura are being laid, even though catfish showed no comparable
10 benefit from the structures. Whether this translates into a detectable population-level effect
11 will depend on continued long-term monitoring — the logical next phase of this restoration
12 programme.

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7 **Authors’ contributions**

8 Study conception and design were undertaken by Olivier V. Raven, Ian A. K. Kusabs, Robin
1 Holmes, Francis J. Burdon and Deniz Özkundakci. Fieldwork was executed by Olivier V. Raven.
2 Data analysis was performed by Olivier V. Raven. Ian A. K. Kusabs obtained council and iwi
3 consent and provided local ecological knowledge. Manuscript drafting was led by Olivier V.
4 Raven. All authors contributed to manuscript revision, approved the results, and the final
1 manuscript.

2 Use of Artificial Intelligence

3 Use of Artificial Intelligence: Claude (Anthropic) was used to assist with manuscript editing and
4 language refinement. All content was critically reviewed, and the authors take full responsibility
5 for the final manuscript.

6 Declarations

7 The corresponding author has declared that none of the authors has any competing interests.

8 Data availability statement

9 The code and derived data supporting the findings of this study are openly available on GitHub
10 (https://github.com/OlivierRaven/Stone_Piles_Rotoiti) and a rendered version of the analysis
1 is hosted at https://olivierraven.github.io/Stone_Piles_Rotoiti/. The primary raw dataset is
2 archived on Zenodo (<https://doi.org/...>). Peer review documents, including the cover letter
3 and reviewer responses, are available in the manuscript repository in the interest of open and
4 transparent science.

5 Tables

Table 1: Physical characteristics of stone pile, control, and reference monitoring sites at baseline. Sites 1–10 are stone pile (odd) and paired control (even) sites; sites 11–20 are reference sites. Substrate index is a weighted composite of substrate particle sizes (higher values indicate coarser substrate more suitable for kōura). Slope is the gradient to the 20 m depth contour. Fetch is maximum effective fetch. Vegetation values are visually estimated percentage cover.

Site	Type	Habitat	Substrate index	Slope (m/m)	Fetch (km)	Riparian veg. (%)	Overhang (%)	Wood cover (%)
1	Stone pile	Rocky	5.20	0.12	4.76	0	10	0
2	Control	Rocky	4.40	0.11	4.79	70	0	0
3	Stone pile	Sandy	3.00	0.01	1.71	100	0	0
4	Control	Sandy	3.00	0.01	1.74	80	5	0
5	Stone pile	Sandy	2.40	0.01	0.61	10	0	0
6	Control	Sandy	3.00	0.01	0.55	0	0	0
7	Stone pile	Sandy	3.00	0.06	2.50	100	0	0
8	Control	Sandy	3.00	0.05	2.22	30	0	0

9	Stone pile	Sandy	3.35	0.08	3.59	0	0	3
10	Control	Rocky	5.00	0.09	3.74	20	0	0
11	Reference	Rocky	6.60	0.16	4.47	0	0	0
12	Reference	Rocky	6.20	0.12	4.17	100	100	0
13	Reference	Sandy	3.60	0.09	1.71	100	100	0
14	Reference	Sandy	3.50	0.11	1.65	100	100	0
15	Reference	Rocky	5.20	0.03	1.44	100	100	6
16	Reference	Rocky	5.20	0.04	1.40	100	100	2
17	Reference	Rocky	5.00	0.05	3.47	100	100	30
18	Reference	Rocky	6.20	0.05	3.46	100	100	7
19	Reference	Rocky	5.70	0.11	2.63	100	100	4
20	Reference	Rocky	5.70	0.13	2.87	100	100	0

6 Source: [Article Notebook](#)

7 Figures

8 Supplementary information

9 *Supplementary figures are provided in the appendices.*

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

Model	RQ	Species	Re- sponse type	Com- pari- son	Time frame	Term	Es- ti- mate	SE	Statis- tic	p	p adj (BH)	95% CI low	95% CI high
M1a	RQ1	Catfish	abundance	treatment	before	(Intercept)	-	0.908	-	0.011	NA	-	-
M1a	RQ1	Catfish	abundance	treatment	before	Site_type_de- tail	2.303	2.535	-	0.652	0.783	4.083	0.522
M1a	RQ1	Catfish	biomass	treatment	before	Stone_pile	0.693	0.452	-	0.012	NA	3.701	5.709
M1a	RQ1	Catfish	biomass	treatment	before	Site_type_de- tail	-	3.107	-	0.101	0.403	-	0.990
M1a	RQ1	Catfish	biomass	treatment	before	Stone_pile	5.100	1.641	-	0.016	NA	11.190	0.358
M1a	RQ1	Kōura	abundance	treatment	before	(Intercept)	-	0.785	-	0.016	NA	-	-
M1a	RQ1	Kōura	abundance	treatment	before	Site_type_de- tail	1.897	2.415	-	0.016	NA	3.437	0.358

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

Model	RQ	Species	Re-	Com-	Time	Term	Es-	Statis-		p	95%	95%	
			sponse	pari-				adj	CI	CI			
			type	son	frame		mate	SE	tic	p	(BH)	low	high
M1a	RQ1	Kōura	abun-	treat-	be-	Site_type_de-	0.288	1.044	0.276	0.783	0.783	-	2.334
			dance	ment	fore	tailStone_pile						1.759	
M1a	RQ1	Kōura	biomass	treat-	be-	(Intercept)	1.030	0.830	1.241	0.215	NA	-	2.656
				ment	fore							0.597	
M1a	RQ1	Kōura	biomass	treat-	be-	Site_type_de-	1.044	0.977	1.068	0.286	0.571	-	2.959
				ment	fore	tailStone_pile						0.872	
M1b	RQ1	Cat-	abun-	ref-	be-	(Intercept)	-	32.656	-	0.002	NA	-	-
		fish	dance	er-	fore		102.299	3.133				166.303	38.295
				ence									
M1b	RQ1	Cat-	abun-	ref-	be-	Site_type_de-	-	0.536	-	0.002	0.004	-	-
		fish	dance	er-	fore	tailReference_2	1.655	3.087				2.706	0.604
				ence									
M1b	RQ1	Cat-	abun-	ref-	be-	Temperature	4.079	1.293	3.154	0.002	NA	1.544	6.613
		fish	dance	er-	fore								
				ence									
M1b	RQ1	Cat-	biomass	ref-	be-	(Intercept)	-	37.363	-	0.000	NA	-	-
		fish		er-	fore		153.136	4.099				226.366	79.906
				ence									
M1b	RQ1	Cat-	biomass	ref-	be-	Site_type_de-	-	0.743	-	0.001	0.002	-	-
		fish		er-	fore	tailReference_2	2.569	3.457				4.026	1.112
				ence									
M1b	RQ1	Cat-	biomass	ref-	be-	Temperature	6.315	1.482	4.261	0.000	NA	3.410	9.220
		fish		er-	fore								
				ence									
M1b	RQ1	Kōura	abun-	ref-	be-	(Intercept)	0.742	0.553	1.343	0.179	NA	-	1.825
			dance	er-	fore							0.341	
				ence									
M1b	RQ1	Kōura	abun-	ref-	be-	Site_type_de-	0.031	0.778	0.040	0.968	0.968	-	1.557
			dance	er-	fore	tailReference_2						1.494	
				ence									

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

Model	Re- sponse	Com- pari- son	Time frame	Term	Es- ti- mate	SE	Statis- tic	p	p adj (BH)	95% CI low	95% CI high
M1b	RQ1	Kourab	biomassref- er- ence	(Intercept)	3.623	0.478	7.580	0.000	NA	2.686	4.560
M1b	RQ1	Kourab	biomassref- er- ence	Site_type_de- tailReference_2	- 0.041	0.681	- 0.061	0.952	0.968	- 1.375	1.293
M2a	RQ2	Cat- fish	abun- dance	vs_confull (Intercept)	- 3.976	1.631	- 2.438	0.015	NA	- 7.173	- 0.780
M2a	RQ2	Cat- fish	abun- dance	vs_confull PeriodAfter	0.673	1.521	0.442	0.658	NA	- 2.308	3.654
M2a	RQ2	Cat- fish	abun- dance	vs_confull siteStone_pile	- 0.639	1.762	- 0.362	0.717	NA	- 4.093	2.816
M2a	RQ2	Cat- fish	abun- dance	vs_confull monitoring_int	- 0.294	0.344	- 0.853	0.394	NA	- 0.968	0.381
M2a	RQ2	Cat- fish	abun- dance	vs_confull fetch_z	- 1.194	1.077	- 1.109	0.267	NA	- 3.305	0.916
M2a	RQ2	Cat- fish	abun- dance	vs_confull Peri- odAfter:site- Stone_pile	2.002	1.973	1.015	0.310	0.858	- 1.864	5.869
M2a	RQ2	Cat- fish	biomassvs- trol	vs_confull (Intercept)	0.202	4.319	0.047	0.963	NA	- 8.262	8.667
M2a	RQ2	Cat- fish	biomassvs- trol	vs_confull PeriodAfter	- 0.340	2.404	- 0.141	0.888	NA	- 5.052	4.372
M2a	RQ2	Cat- fish	biomassvs- trol	vs_confull siteStone_pile	- 5.574	7.158	- 0.779	0.436	NA	- 19.604	8.456
M2a	RQ2	Cat- fish	biomassvs- trol	vs_confull monitoring_int	- 0.381	0.523	- 0.728	0.466	NA	- 1.406	0.644
M2a	RQ2	Cat- fish	biomassvs- trol	vs_confull fetch_z	- 1.675	2.438	- 0.687	0.492	NA	- 6.454	3.103

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

Model	Re- sponse Species	Com- pari- son type	Time frame	Term	Es- ti- mate	SE	Statis- tic	p	p adj (BH)	95% CI low	95% CI high
M2aRQ26	Cat- fish	biomassvs_ confull trol		Peri- odAfter:site- Stone_pile (Intercept)	5.436	5.592	0.972	0.331	0.858	- 5.524	16.396
M2aRQ26	Cat- fish	occu- pancy trol		PeriodAfter	- 2.185	1.887 1.158	- 0.247	NA		- 5.883	1.513
M2aRQ26	Cat- fish	occu- pancy trol		siteStone_pile	- 0.161	1.765 0.091	- 0.927	NA		- 3.621	3.299
M2aRQ26	Cat- fish	occu- pancy trol		monitoring_int	0.024	1.982	0.012	0.990	NA	- 3.860	3.909
M2aRQ26	Cat- fish	occu- pancy trol		fetch_z	- 0.165	0.259 0.635	- 0.525	NA		- 0.673	0.343
M2aRQ26	Cat- fish	occu- pancy trol		Peri- odAfter:site- Stone_pile (Intercept)	- 0.387	1.197 0.802	- 0.423	NA		- 3.306	1.387
M2aRQ26	Cat- fish	occu- pancy trol		PeriodAfter	0.387	2.168	0.179	0.858	0.858	- 3.863	4.637
M2aRQ26	Cat- fish	occu- pancy trol		siteStone_pile	- 0.751	1.183 2.327	- 0.020	NA		- 5.072	- 0.434
M2aRQ26	Cat- fish	occu- pancy trol		monitoring_int	2.374	0.976	2.432	0.015	NA	0.461	4.286
M2aRQ26	Cat- fish	occu- pancy trol		PeriodAfter	0.751	1.335	0.563	0.574	NA	- 1.865	3.368
M2aRQ26	Cat- fish	occu- pancy trol		Stone_pile	- 0.253	0.082 3.094	- 0.002	NA		- 0.413	- 0.093
M2aRQ26	Cat- fish	occu- pancy trol		PeriodAfter:site- Stone_pile (Intercept)	0.311	1.232	0.253	0.800	0.858	- 2.104	2.727
M2aRQ26	Cat- fish	occu- pancy trol		biomassvs_ confull trol	0.315	1.335	0.236	0.813	NA	- 2.301	2.931

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

Model	Re- sponse	Com- pari- son	Time frame	Term	Es- ti- mate	SE	Statis- tic	p	p adj (BH)	95% CI low	95% CI high
M2aBQ2Kōura	biomass	vs_confull	trol	PeriodAfter	2.248	1.246	1.805	0.071	NA	-	4.689
M2aBQ2Kōura	biomass	vs_confull	trol	siteStone_pile	1.410	1.552	0.909	0.363	NA	-	4.451
M2aBQ2Kōura	biomass	vs_confull	trol	monitoring_int	-	0.110	-	0.073	NA	-	0.019
M2aBQ2Kōura	biomass	vs_confull	trol	PeriodAfter:site-	-	1.471	-	0.819	0.858	-	2.548
M2aBQ2Kōura	biomass	vs_confull	trol	Stone_pile	0.336	0.228				3.220	
M2aBQ2Kōura	occu- pancy	vs_confull	trol	(Intercept)	-	1.643	-	0.674	NA	-	2.530
M2aBQ2Kōura	occu- pancy	vs_confull	trol	PeriodAfter	1.913	1.556	1.230	0.219	NA	-	4.963
M2aBQ2Kōura	occu- pancy	vs_confull	trol	siteStone_pile	0.013	2.121	0.006	0.995	NA	-	4.171
M2aBQ2Kōura	occu- pancy	vs_confull	trol	monitoring_int	-	0.215	-	0.183	NA	-	0.135
M2aBQ2Kōura	occu- pancy	vs_confull	trol	PeriodAfter:site-	1.013	1.762	0.575	0.565	0.858	-	4.466
M2aBQ2Kōura	occu- pancy	vs_confull	trol	Stone_pile						2.440	
M2bBQ2C	abun- fish	vs_conaf-	trol	(Intercept)	-	1.519	-	0.157	NA	-	0.826
M2bBQ2C	abun- fish	vs_conaf-	trol	siteStone_pile	2.150	1.416				5.127	
M2bBQ2C	abun- fish	vs_conaf-	trol	monitoring_int	-	0.406	-	0.118	NA	-	0.162
M2bBQ2C	abun- fish	vs_conaf-	trol	fetch_z	-	0.976	-	0.261	NA	-	0.816
M2bBQ2C	abun- fish	vs_conaf-	trol		1.097	1.124				3.010	

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

Model	Re- sponse	Com- pari- son	Time frame	Term	Es- ti- mate	SE	Statis- tic	p	p adj (BH)	95% CI low	95% CI high
M2b	Cat- fish	abun- dance	vs_conaf- trol	site- Stone_pile:mon- itoring_int (Intercept)	0.831	0.557	1.492	0.136	0.215	-	1.922
M2b	Cat- fish	biomass	vs_conaf- trol	site- Stone_pile	2.964	2.725	1.088	0.277	NA	-	8.304
M2b	Cat- fish	biomass	vs_conaf- trol	monitoring_int	-	4.240	-	0.286	NA	-	3.789
M2b	Cat- fish	biomass	vs_conaf- trol	fetch_z	4.522	1.066				12.833	
M2b	Cat- fish	biomass	vs_conaf- trol	site- Stone_pile:mon- itoring_int	-	0.688	-	0.157	NA	-	0.374
M2b	Cat- fish	biomass	vs_conaf- trol	(Intercept)	0.974	1.416				2.323	
M2b	Cat- fish	biomass	vs_conaf- trol	fetch_z	-	1.390	-	0.524	NA	-	1.837
M2b	Cat- fish	biomass	vs_conaf- trol	site- Stone_pile:mon- itoring_int	0.886	0.638				3.610	
M2b	Cat- fish	biomass	vs_conaf- trol	(Intercept)	1.357	0.974	1.393	0.164	0.215	-	3.266
M2b	Cat- fish	occu- pancy	vs_conaf- trol	site- Stone_pile:mon- itoring_int	-	1.737	-	0.584	NA	-	2.453
M2b	Cat- fish	occu- pancy	vs_conaf- trol	monitoring_int	0.952	0.548				4.357	
M2b	Cat- fish	occu- pancy	vs_conaf- trol	fetch_z	-	1.936	-	0.298	NA	-	1.781
M2b	Cat- fish	occu- pancy	vs_conaf- trol	(Intercept)	2.013	1.040				5.808	
M2b	Cat- fish	occu- pancy	vs_conaf- trol	site- Stone_pile:mon- itoring_int	-	0.420	-	0.179	NA	-	0.258
M2b	Cat- fish	occu- pancy	vs_conaf- trol	monitoring_int	0.565	1.345				1.388	
M2b	Cat- fish	occu- pancy	vs_conaf- trol	fetch_z	-	1.121	-	0.444	NA	-	1.339
M2b	Cat- fish	occu- pancy	vs_conaf- trol	(Intercept)	0.858	0.766				3.055	
M2b	Cat- fish	occu- pancy	vs_conaf- trol	site- Stone_pile:mon- itoring_int	0.735	0.546	1.344	0.179	0.215	-	1.806
M2b	Cat- fish	occu- pancy	vs_conaf- trol	monitoring_int						0.336	
M2b	Kōura	abun- dance	vs_conaf- trol	(Intercept)	0.084	0.846	0.099	0.921	NA	-	1.741
M2b	Kōura	abun- dance	vs_conaf- trol	site- Stone_pile						1.573	
M2b	Kōura	abun- dance	vs_conaf- trol	monitoring_int	0.323	0.808	0.400	0.689	NA	-	1.907
M2b	Kōura	abun- dance	vs_conaf- trol	fetch_z						1.261	

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

Model	Re- sponse Species	Com- pari- son type	Time frame	Term	Es- ti- mate	SE	Statis- tic	p	p adj (BH)	95% CI low	95% CI high
M2b	Kōura	abun- dance	vs_conaf- trol	monitoring_int	-	0.143	-	0.004	NA	-	-
M2b	Kōura	abun- dance	vs_conaf- trol	site-	0.243	0.179	1.359	0.174	0.215	-	0.593
M2b	Kōura	abun- dance	vs_conaf- trol	Stone_pile:mon- itoring_int						0.107	
M2b	Kōura	biomass	vs_conaf- trol	(Intercept)	3.270	0.856	3.821	0.000	NA	1.593	4.948
M2b	Kōura	biomass	vs_conaf- trol	siteStone_pile	-	0.986	-	0.938	NA	-	1.856
M2b	Kōura	biomass	vs_conaf- trol	monitoring_int	-	0.191	-	0.025	NA	-	-
M2b	Kōura	biomass	vs_conaf- trol	site-	0.366	0.239	1.531	0.126	0.215	-	0.834
M2b	Kōura	biomass	vs_conaf- trol	Stone_pile:mon- itoring_int						0.103	
M2b	Kōura	occu- pancy	vs_conaf- trol	(Intercept)	1.585	1.885	0.841	0.400	NA	-	5.280
M2b	Kōura	occu- pancy	vs_conaf- trol	siteStone_pile	0.563	2.660	0.212	0.832	NA	-	5.776
M2b	Kōura	occu- pancy	vs_conaf- trol	monitoring_int	-	0.311	-	0.195	NA	-	0.207
M2b	Kōura	occu- pancy	vs_conaf- trol	site-	0.229	0.441	0.519	0.604	0.604	-	1.094
M2b	Kōura	occu- pancy	vs_conaf- trol	Stone_pile:mon- itoring_int						0.636	
M3a	Cat- fish	abun- dance	vs_ref-af- er- ence	(Intercept)	-	1.282	-	0.000	NA	-	-
M3a	Cat- fish	abun- dance	vs_ref-af- er- ence	siteStone_pile	-	1.940	-	0.203	NA	-	1.331
M3a	Cat- fish	abun- dance	vs_ref-af- er- ence		2.471	1.274				6.273	

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

Model	Re- sponse Type	Com- pari- son	Time frame	Term	Es- ti- mate	SE	Statis- tic	p	p adj (BH)	95% CI low	95% CI high
M3a	RQ36	Cat- fish	abun- dance	vs_ref-af- er-ter ence	monitoring_int	- 0.215	0.147 1.462	- 0.144	NA	- 0.504	0.073
M3a	RQ36	Cat- fish	abun- dance	vs_ref-af- er-ter ence	Temperature	0.315	0.0585.414	0.000	NA	0.201	0.429
M3a	RQ36	Cat- fish	abun- dance	vs_ref-af- er-ter ence	site- Stone_pile:mon- itoring_int	0.510	0.3981.283	0.199	0.559	- 0.269	1.290
M3a	RQ36	Cat- fish	biomass	vs_ref-af- er-ter ence	(Intercept)	3.011	1.1892.532	0.011	NA	0.681	5.342
M3a	RQ36	Cat- fish	biomass	vs_ref-af- er-ter ence	siteStone_pile	- 4.370	3.059 1.429	- 0.153	NA	- 10.366	1.626
M3a	RQ36	Cat- fish	biomass	vs_ref-af- er-ter ence	monitoring_int	- 0.250	0.209 1.196	- 0.232	NA	- 0.660	0.160
M3a	RQ36	Cat- fish	biomass	vs_ref-af- er-ter ence	site- Stone_pile:mon- itoring_int	0.689	0.6381.081	0.280	0.559	- 0.560	1.939
M3a	RQ36	Kōura	abun- dance	vs_ref-af- er-ter ence	(Intercept)	0.229	0.5190.440	0.660	NA	- 0.789	1.247
M3a	RQ36	Kōura	abun- dance	vs_ref-af- er-ter ence	siteStone_pile	0.217	0.9010.240	0.810	NA	- 1.550	1.984
M3a	RQ36	Kōura	abun- dance	vs_ref-af- er-ter ence	monitoring_int	- 0.090	0.056 1.612	- 0.107	NA	- 0.199	0.019

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

Model	Re- sponse Type	Com- pari- son	Time frame	Term	Es- ti- mate	SE	Statis- tic	p	p adj (BH)	95% CI low	95% CI high
M3aRQ3K	Kōura	abun- dance	vs_ref- er- ence	af- ter site- Stone_pile:mon- itoring_int (Intercept)	- 0.086 3.369	0.109 0.506	- 0.791 6.655	0.429 0.000	0.572 NA	- 0.299 2.377	0.127 4.361
M3aRQ3K	Kōura	biomass	vs_ref- er- ence	af- ter siteStone_pile	- 0.149	0.885 0.168	- 0.866	NA	- 1.884	- 1.586	1.586
M3aRQ3K	Kōura	biomass	vs_ref- er- ence	af- ter monitoring_int	- 0.088	0.074 1.177	- 0.239	NA	- 0.234	- 0.058	0.058
M3aRQ3K	Kōura	biomass	vs_ref- er- ence	af- ter site- Stone_pile:mon- itoring_int (Intercept)	0.016 7.372	0.142 4.324	2.112 0.911	0.911 0.000	NA	- 10.713	0.294 4.031
M3bRQ3C	Cat- fish	abun- dance	ref- er- ence	af- ter Site_type_de- tailReference_2 (Intercept)	0.753 1.007	1.007 0.747	0.455 0.455	NA	- 1.221	- 2.726	- 2.726
M3bRQ3C	Cat- fish	abun- dance	ref- er- ence	af- ter monitoring_int	- 0.130	0.220 0.591	- 0.555	NA	- 0.562	- 0.301	0.301
M3bRQ3C	Cat- fish	abun- dance	ref- er- ence	af- ter Temperature	0.304	0.073	4.183	0.000	NA	0.161	0.446

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

Model	Re- sponse Type	Com- pari- son	Time frame	Term	Es- ti- mate	SE	Statis- tic	p	p adj (BH)	95% CI low	95% CI high
M3b	RQ36	Cat- fish	abun- dance	ref- er- ence	af- ter	Site_type_de- tailRefer- ence_2:moni- toring_int (Intercept)	- 0.151	0.287 0.527	- 0.598	0.797 0.714	- 0.411
M3b	RQ36	Cat- fish	biomass	ref- er- ence	af- ter	(Intercept)	2.726	1.556	1.752	0.080	NA - 5.775 0.324
M3b	RQ36	Cat- fish	biomass	ref- er- ence	af- ter	Site_type_de- tailReference_2	0.191	1.640	0.116	0.907	NA - 3.406 3.024
M3b	RQ36	Cat- fish	biomass	ref- er- ence	af- ter	monitoring_int	- 0.227	0.301 0.754	- 0.451	NA 0.816	- 0.363
M3b	RQ36	Cat- fish	biomass	ref- er- ence	af- ter	Site_type_de- tailRefer- ence_2:moni- toring_int (Intercept)	0.022	0.419	0.052	0.959	0.959 - 0.843 0.800
M3b	RQ36	Kōura	abun- dance	ref- er- ence	af- ter	(Intercept)	0.380	0.790	0.480	0.631	NA - 1.928 1.169
M3b	RQ36	Kōura	abun- dance	ref- er- ence	af- ter	Site_type_de- tailReference_2	- 0.115	0.570 0.202	- 0.840	NA 1.232	- 1.002
M3b	RQ36	Kōura	abun- dance	ref- er- ence	af- ter	monitoring_int	- 0.019	0.062 0.302	- 0.763	NA 0.141	- 0.103
M3b	RQ36	Kōura	abun- dance	ref- er- ence	af- ter	SeasonSpring	- 0.386	0.227 1.699	- 0.089	NA 0.831	- 0.059

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

Model	Re- sponse type	Com- pari- son	Time frame	Term	Es- ti- mate	SE	Statis- tic	p	p adj (BH)	95% CI low	95% CI high
M3b	RQ3K	Kōura	abun- dance	ref- er- ence	af- ter	SeasonSummer	- 0.233 0.330	- 1.416	0.157 NA	- 0.787	0.127
M3b	RQ3K	Kōura	abun- dance	ref- er- ence	af- ter	SeasonWinter	- 0.196 0.546	- 2.781	0.005 NA	- 0.932	- 0.161
M3b	RQ3K	Kōura	abun- dance	ref- er- ence	af- ter	Site_type_de- tailRefer- ence_2:moni- toring_int	- 0.088 0.069	- 0.787	0.4310.797	- 0.241	0.103
M3b	RQ3K	Kōura	biomass	ref- er- ence	af- ter	(Intercept)	3.275 0.800	4.095 0.000	NA	1.707	4.842
M3b	RQ3K	Kōura	biomass	ref- er- ence	af- ter	Site_type_de- tailReference_2	- 0.670 0.067	- 0.100	0.920 NA	- 1.379	1.245
M3b	RQ3K	Kōura	biomass	ref- er- ence	af- ter	monitoring_int	- 0.087 0.047	- 0.548	0.584 NA	- 0.217	0.122
M3b	RQ3K	Kōura	biomass	ref- er- ence	af- ter	Site_type_de- tailRefer- ence_2:moni- toring_int	- 0.125 0.074	- 0.591	0.5540.797	- 0.319	0.171



Figure 1: Map of Lake Rotoiti in Bay of Plenty region of New Zealand. The monitoring sites are coloured by site type: stone pile sites red, control site yellow, and reference sites blue. Ten close-up maps show the placement of each of the sites.

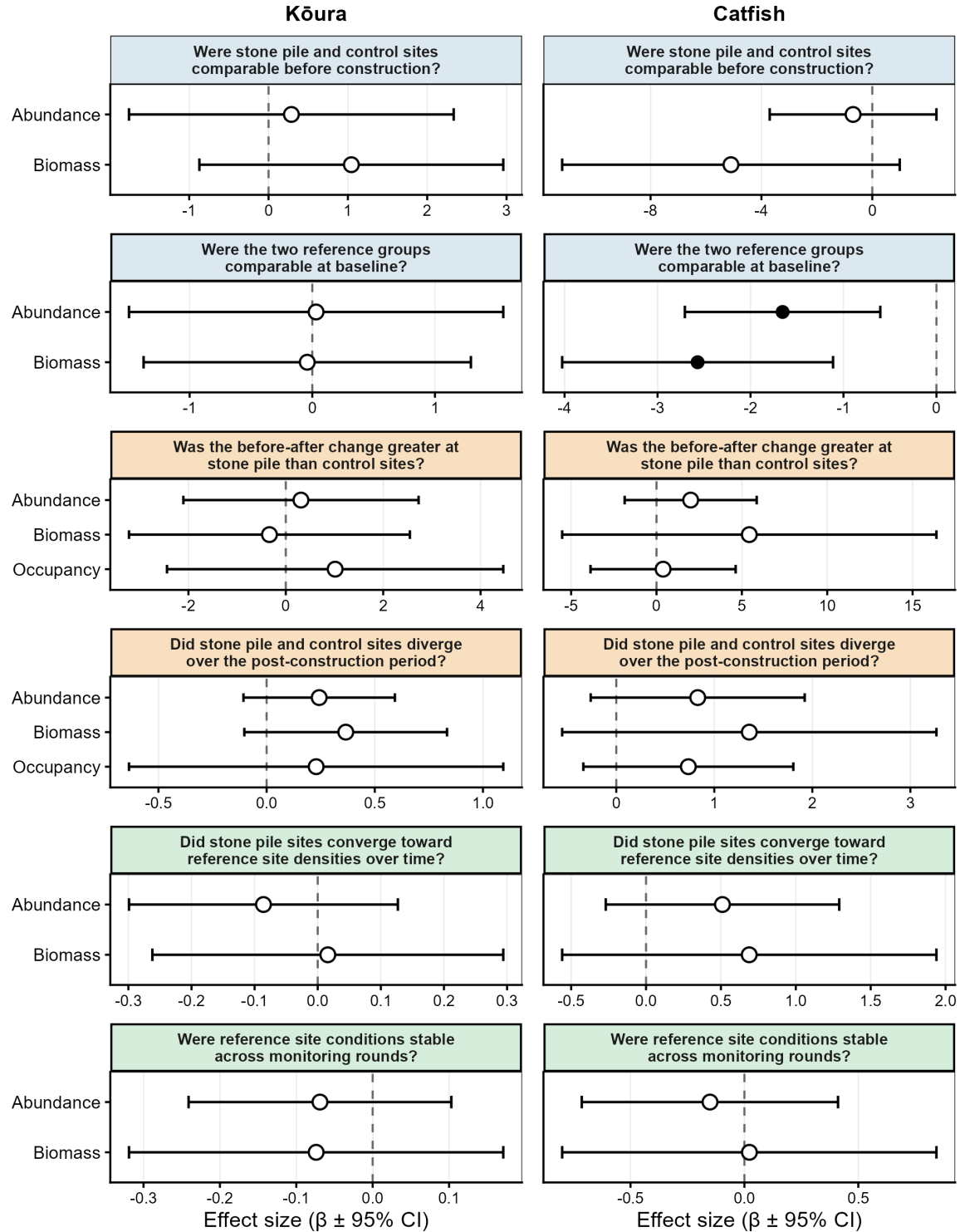


Figure 2: Effect sizes ($\beta \pm 95\% \text{ CI}$) for the key term of interest across all models, separated by species (columns) and comparison type (rows). Filled points indicate significant effects ($p < 0.05$); open points are non-significant. Each row has its own x-axis scale, effect magnitudes are not directly comparable across rows because each comparison tests a different contrast (step-change, per-round slope, or simple group difference), and occupancy is on the log-odds scale while abundance and biomass models use the Tweedie log link. Models with degenerate confidence intervals are excluded.

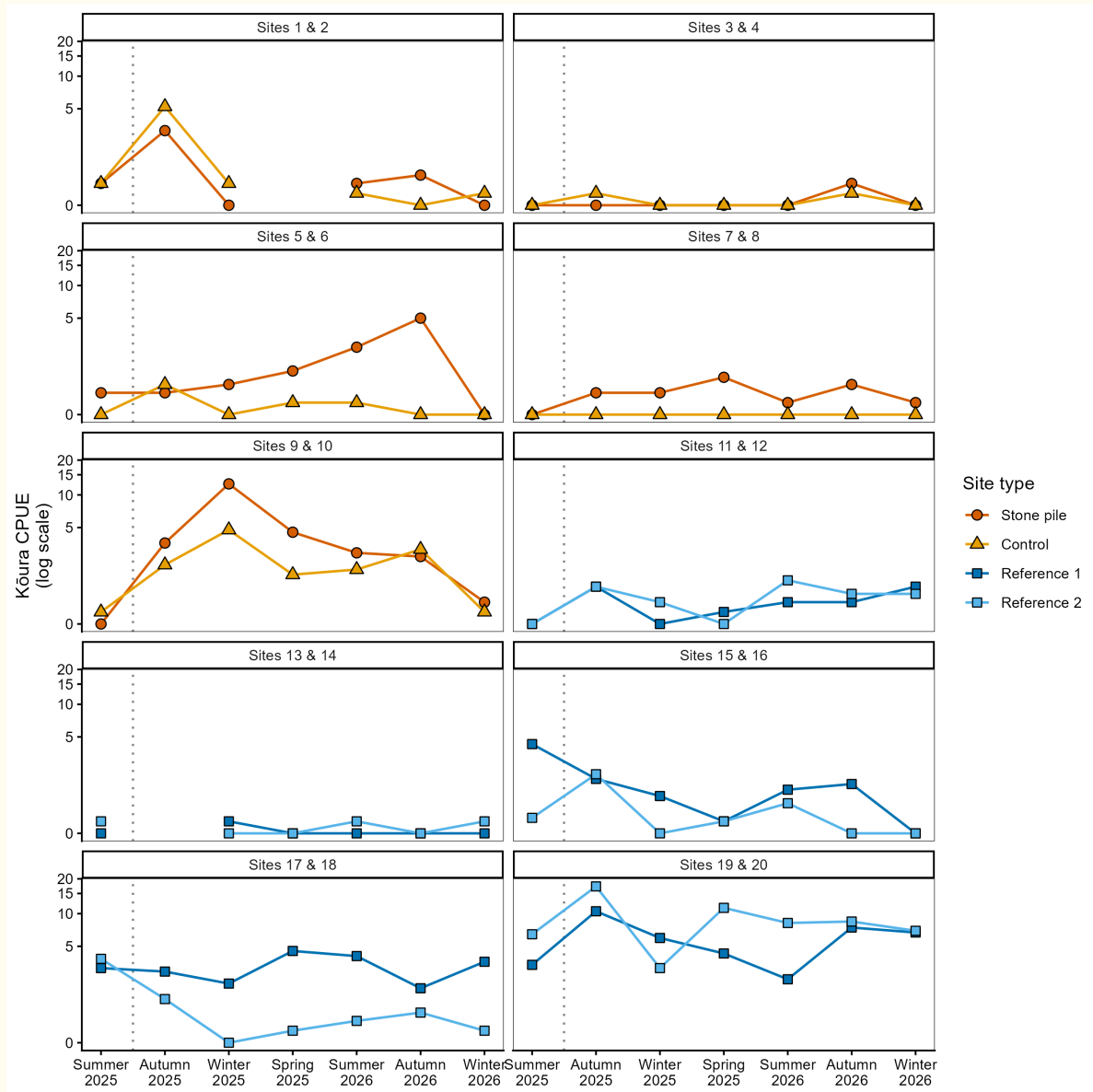


Figure 3: Kōura CPUE at individual site pairs across all monitoring rounds. Each panel shows one stone pile / control pair (sites 1–10) or reference pair (sites 11–20). Colour and shape indicate site type. Vertical dotted line marks stone pile construction. y-axis on $\log(1+x)$ scale.

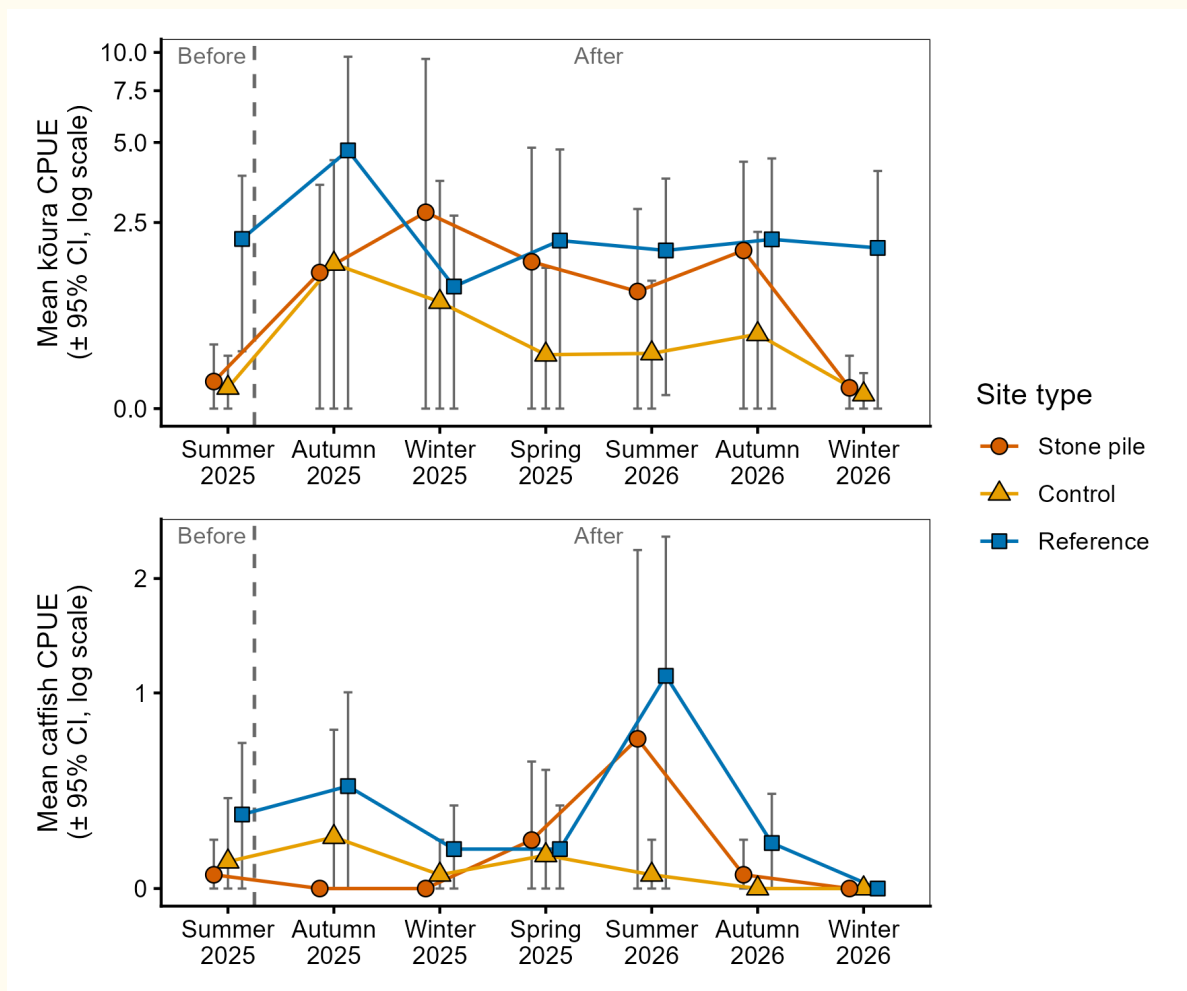


Figure 4: Mean CPUE per monitoring round ($\pm 95\%$ CI, log scale) by site type for kōura (top) and catfish (bottom). Vertical dashed line distinguishes the pre-construction period (Before) from post-construction monitoring (After).

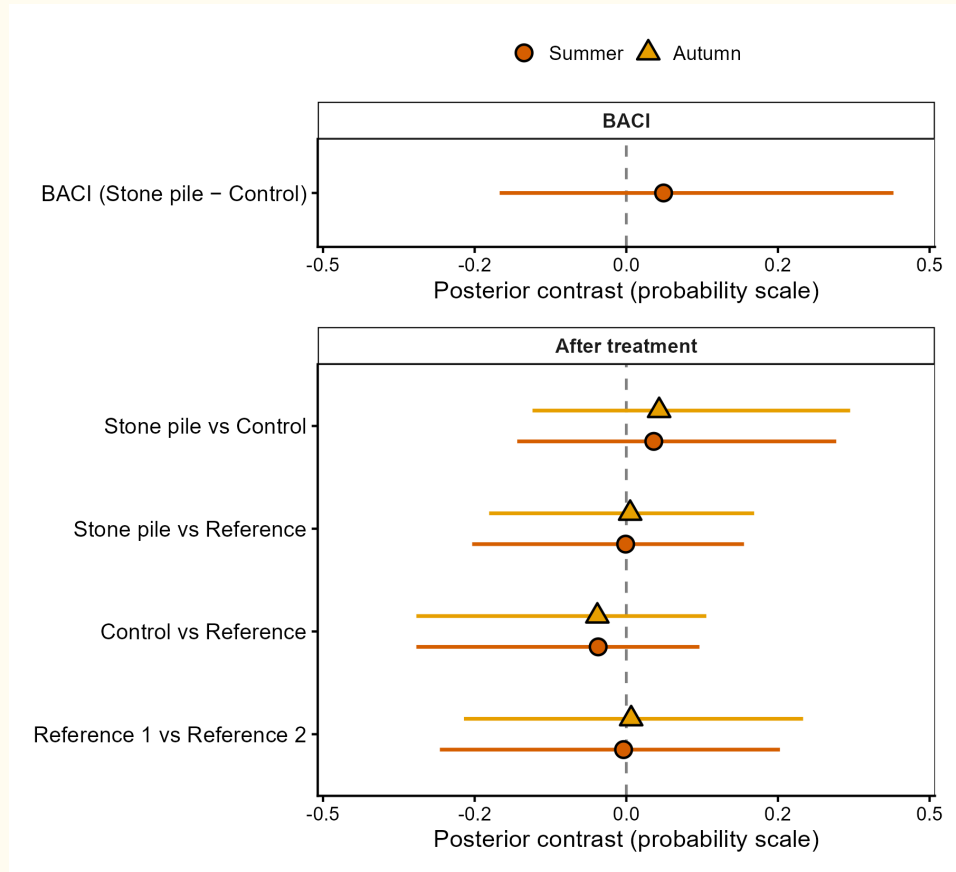


Figure 5: Posterior contrasts in kōura occupancy from the warm-season model. BACI contrast (stone pile minus control, before vs after construction). After-treatment contrasts between all site-type pairs. Points show posterior means with 95 % credible intervals; contrasts overlapping zero indicate no detectable difference.

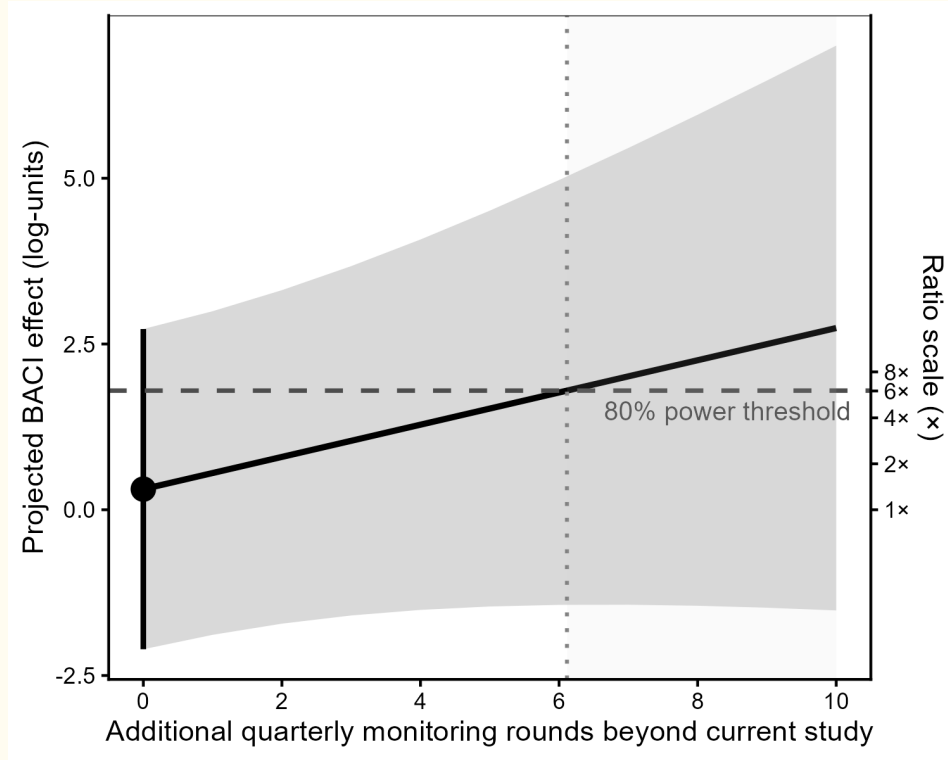


Figure 6: Projected kōura CPUE BACI effect size over additional monitoring rounds. The solid line shows the linear projection based on the estimated post-construction trajectory; shaded ribbon shows 95% uncertainty interval. Dashed horizontal line marks the detection threshold for 80 % power with five site pairs.

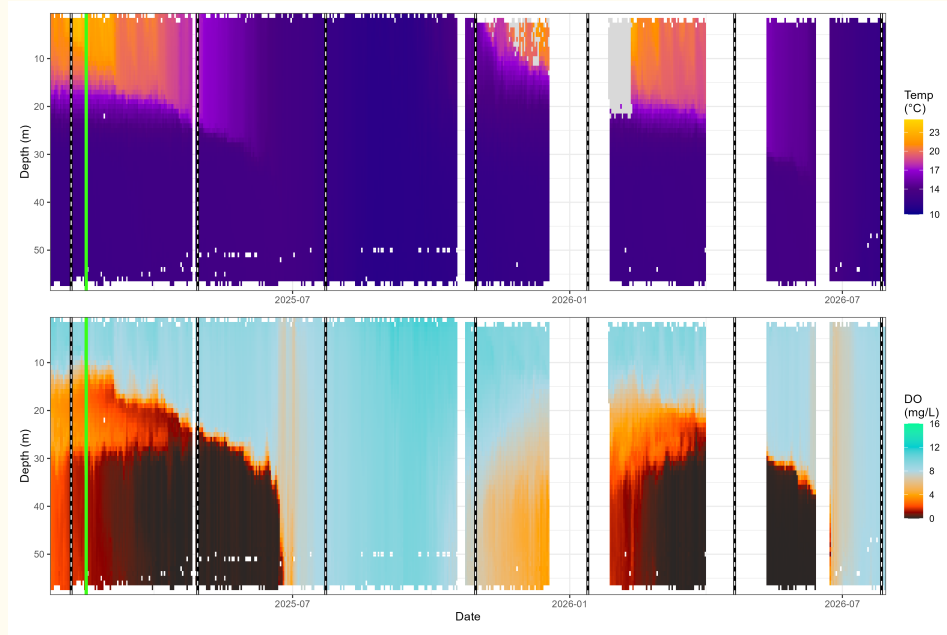


Figure 7: Depth-resolved water temperature ($^{\circ}\text{C}$, top) and dissolved oxygen (mg/L , bottom) in Lake Rotoiti throughout the study period (February 2025 – July 2026). Data derived from the LimnoTrack continuous monitoring buoy ([Moore & McBride, 2026](#)). Vertical green line marks stone pile deployment; dashed vertical lines mark seasonal monitoring rounds. Warm colours in the DO panel (orange–black) indicate hypoxic and anoxic conditions ($< 5 \text{ mg/L}$), demonstrating the pronounced summer stratification that compresses oxygenated habitat into the shallow littoral zone.

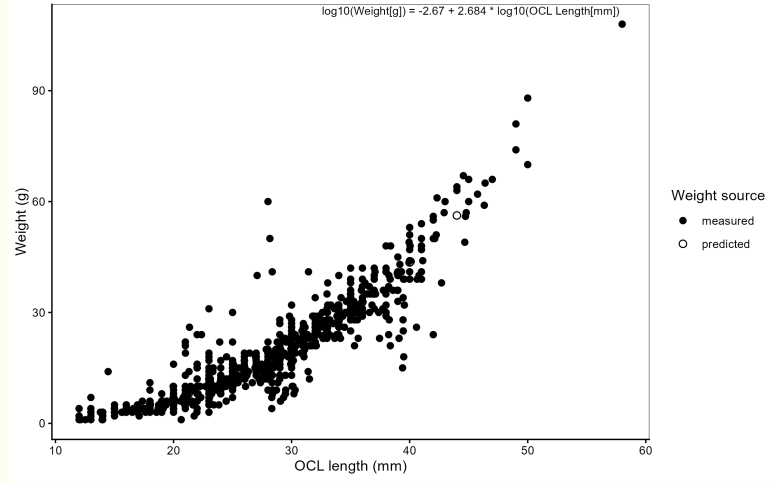


Figure 8: Length–weight relationship for kōura, fitted on log –log scale. Open points are weights predicted from the length–weight regression (used to fill missing scale measurements); filled points are directly measured weights.

Table 3: Baseline (pre-construction) comparability of stone pile, control, and reference sites. Values are mean $\pm$ SD; p-values test Stone pile vs Control (paired Wilcoxon signed-rank test) and Stone pile vs Reference (unpaired Wilcoxon rank-sum test).

Variable	Stone pile	Control	Reference	p vs Control	p vs Reference
Kōura CPUE	0.2 ± 0.27	0.15 ± 0.22	2.13 ± 2.32	1.000	0.205
Catfish CPUE	0.05 ± 0.11	0.1 ± 0.22	0.3 ± 0.52	1.000	0.420
Kōura BPUE	7.95 ± 12.15	2.8 ± 3.84	36.69 ± 36.82	0.789	0.206
Catfish BPUE	0.15 ± 0.34	24.6 ± 55.01	97.28 ± 232.15	1.000	0.343
Substrate index (before)	3.39 ± 1.07	3.68 ± 0.95	5.29 ± 1.05	0.789	0.012
Substrate index (after)	3.65 ± 1.12	3.68 ± 0.95	5.29 ± 1.05	1.000	0.032
Slope (0–20 m)	0.06 ± 0.05	0.06 ± 0.04	0.09 ± 0.05	0.590	0.245

Table 3: Baseline (pre-construction) comparability of stone pile, control, and reference sites. Values are mean $\pm$ SD; p-values test Stone pile vs Control (paired Wilcoxon signed-rank test) and Stone pile vs Reference (unpaired Wilcoxon rank-sum test).

Variable	Stone pile	Control	Reference	p vs	p vs
				Control	Reference
Temperature ($^{\circ}\text{C}$)	24.74 $\pm$ 1.92	24.78 $\pm$ 1.47	24.48 $\pm$ 0.78	1.000	0.623
Dissolved oxygen (mg/L)	9.85 $\pm$ 1.48	9.63 $\pm$ 1.29	10.03 $\pm$ 0.86	0.418	0.426
Specific conductivity ($\mu\text{S}/\text{cm}$)	195.44 $\pm$ 70.05	199.8 $\pm$ 78.52	165.45 $\pm$ 8.5	0.281	0.951
pH	7.5 $\pm$ 2.15	7.61 $\pm$ 2.2	8.53 $\pm$ 0.66	1.000	0.426

Table 4: Posterior kōura occupancy probability (ψ) from the warm-season occupancy model (Summer and Autumn rounds). Values are population-level estimates averaged over random site effects, with 95 % credible intervals. Autumn has no Before period as baseline monitoring was conducted in summer only.

Season	Period	Site type	(mean [95% CrI])
Summer	Before	Stone pile	0.81 [0.26–1.00]
Summer	Before	Control	0.82 [0.34–1.00]
Summer	Before	Reference 1	0.86 [0.36–1.00]
Summer	Before	Reference 2	0.89 [0.49–1.00]
Summer	After	Stone pile	0.93 [0.64–1.00]
Summer	After	Control	0.89 [0.53–1.00]
Summer	After	Reference 1	0.93 [0.62–1.00]
Summer	After	Reference 2	0.94 [0.67–1.00]
Autumn	After	Stone pile	0.93 [0.69–1.00]
Autumn	After	Control	0.88 [0.53–1.00]
Autumn	After	Reference 1	0.93 [0.65–1.00]
Autumn	After	Reference 2	0.92 [0.63–1.00]

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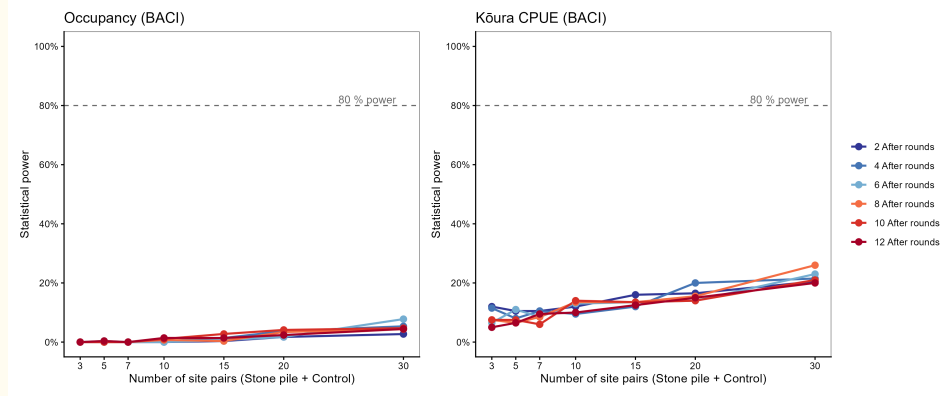


Figure 9: Simulation-based power curves for the occupancy BACI (left) and kōura CPUE BACI (right). Each line represents a different number of After-period monitoring rounds. Dashed line marks 80 % power.

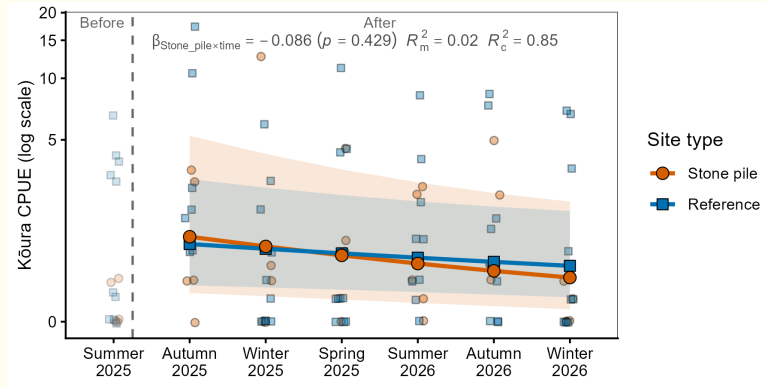


Figure 10: Predicted kōura CPUE trajectories over monitoring rounds at stone pile and reference sites from the abundance convergence model. Shaded ribbons represent 95% confidence intervals. The annotation shows the key interaction term ($\beta \pm p\text{-value}$) and model R^2 values for fixed effects (R^2_m) and conditional (R^2_c).

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